

Somil Doshi

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Summary

Results-driven Data Science graduate with hands-on experience in data engineering, analytics, and predictive modeling across industry and research settings. Skilled in building end-to-end pipelines, designing stakeholder-facing dashboards, and applying machine learning and statistical techniques to deliver actionable insights. Thrives in collaborative, cross-functional environments with a strong focus on translating complex data into clear decisions.

Technical Skills

Languages & Libraries: Python, SQL, R, Pandas, NumPy, Scikit-learn, Streamlit, Flask, Git, MS Excel
Data Engineering & Deployment: Airflow, dbt, Spark, Docker, Kubernetes, GitHub Actions, ETL Pipelines, AWS SageMaker
Databases & Visualization: PostgreSQL, AWS RDS, MongoDB Atlas, MySQL, Power BI, Tableau
Data Science & Modeling: Predictive Modeling, Feature Engineering, Statistical Analysis, Data Quality Checks, JSON, Parquet

Professional Experience

University of Houston

March 2025 - December 2025

Research Assistant, Data Engineering and Analytics

- Designed and maintained end-to-end data pipelines for research datasets, supporting ingestion, transformation, and structured storage, improving data availability and analysis readiness by 30%.
- Built and optimized SQL-based analytical data models using **Databricks**, reducing query latency by 22%; collaborated with faculty and peer researchers to align pipeline outputs with evolving research requirements.
- Deployed and managed data services using **PostgreSQL** and **Docker**, ensuring scalable, secure dataset access across multiple concurrent research projects for cross-functional academic teams.
- Conducted exploratory data analysis and statistical reporting to surface trends, presenting findings to research stakeholders and translating results into actionable next steps.

Colin Motors

April 2023 - March 2024

Data Analyst Intern

- Analyzed and consolidated 10K+ motor production, quality, and sales records using **MongoDB** and **Databricks**, orchestrated with **Airflow** and standardized via **dbt**, improving forecasting reliability by 19%.
- Developed time-series demand forecasting pipelines in **Databricks**, achieving 92% forecast accuracy and reducing prediction error by 18% against baseline models; partnered with operations teams to validate outputs and refine model assumptions.
- Designed **Power BI** dashboards with **DAX** to track motor performance KPIs and sales seasonality, communicating insights to commercial and operations stakeholders to support capacity planning decisions.

Projects

RevPAR Intelligence Platform - BroadVail Capital | *Python, Streamlit, OpenAI API, XGBoost, LightGBM, SHAP*

- Built a revenue forecasting pipeline on 12,800+ multifamily property records, achieving 70.6% R^2 and 0.109 RMSE, reducing prediction error by 18% over baseline models using feature engineering and ensemble methods.
- Automated executive reporting using LLM-driven summaries and SHAP-based explainability; collaborated with client stakeholders to refine dashboard outputs, delivering interpretable investment insights via interactive Streamlit interfaces.

Predictive Maintenance Analytics for Oil Wells | *Neural Networks, PostgreSQL, Docker, Streamlit*

- Engineered a sensor data pipeline for drilling operations, applying predictive modeling to improve maintenance lead time by 7 days through early fault detection and anomaly identification.
- Reduced equipment downtime by 28% via data-driven maintenance scheduling across a synthetic dataset modeling 5,000+ well operations; built and documented a Streamlit monitoring dashboard for team use.

Brain Tumor MRI Classification | *3D CNNs, PyTorch, Flask APIs, MongoDB Atlas, Docker*

- Implemented an end-to-end inference pipeline for 3D MRI data, achieving 94% classification accuracy across multiple tumor subtypes through iterative model optimization and rigorous data preprocessing.
- Served predictions via **Flask** REST APIs with **MongoDB Atlas** logging, reducing diagnostic reporting time by 30% and enabling structured storage of results for downstream clinical analysis.

Education

University of Houston

Master of Science in Engineering Data Science

August 2024 - May 2026

GPA: 3.89/4.0

University of Mumbai

Bachelor of Engineering in Information Technology

August 2020 - May 2024

GPA: 3.84/4.0